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Inner ear transplantation of cochlear progenitor cells restores acute sensorineural hearing loss

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ABSTRACT

To investigate whether cochlear progenitor cell transplantation into the inner ear can restore kanamycin-induced acute ototoxic sensorineural hearing loss, we established an acute ototoxic injury model via intraperitoneal injection of kanamycin and a diuretic. Cell transplantation was performed on the fifth day after model establishment. Two weeks post-transplantation, auditory brainstem response (ABR) testing assessed hearing recovery in mice. Subsequently, immunofluorescence staining of mouse cochleae analyzed the distribution and engraftment of transplanted cochlear stem cells within the inner ear. Results demonstrated that kanamycin caused acute loss of outer hair cells while significantly elevating hearing thresholds. Following cell transplantation, cochlear progenitor cells predominantly distributed in the scala tympani beneath the basilar membrane, with a small number colonizing the scala media. Concurrently, hearing thresholds partially recovered in transplanted mice. Collectively, these findings demonstrate that transplanted cochlear progenitor cells can survive and colonize within the cochlea, partially restoring hearing in mice with acute ototoxic damage.

Introduction

Sensorineural hearing loss (SNHL) is one of the most common types of hearing impairment encountered clinically (Yan et al., 2025). Its primary pathological basis involves damage or loss of hair cells and spiral ganglion neurons in the inner ear (Géléoc and Holt, 2014; Li et al., 2025). In mammals, hair cells in the cochlea lack the ability to regenerate. Therefore, damage to these cells results in irreversible hearing loss (Lahlou et al., 2024; Shu et al., 2019). Ototoxic drugs, such as aminoglycoside antibiotics (e.g., kanamycin), are a major cause of acquired sensorineural hearing loss. These drugs severely impair auditory function by promoting hair cell apoptosis or oxidative stress responses (Jiang et al., 2006; Kawamoto et al., 2004; Esterberg et al., 2016; He et al., 2020). Currently, there are numerous treatment options for hearing loss. Commonly used clinical approaches include hearing aid fitting, cochlear implant surgery, glucocorticoid therapy, and hyperbaric oxygen therapy (Schreiber et al., 2010; Rauch et al., 2011). Although these methods can compensate for or restore patients' hearing function to some extent, they also present various challenges. Some involve invasive surgery, while others carry significant toxic side effects. Consequently, these approaches face difficulties in widespread clinical application, underscoring the urgent need to develop safer and more effective treatment strategies (Joshua et al., 2022; Xu et al., 2023; Cederroth et al., 2024).

In recent years, cell therapy has emerged as a highly promising regenerative medicine approach, demonstrating tremendous potential in repairing tissue and organ damage (Karim Rony and Tompkins, 2025; Chen et al., 2025; Brignier and Gewirtz, 2010). However, the application of cell therapy in hearing restoration research remains in its early stages (Lee and Park, 2018; Roccio, 2021). Progenitor cells in the inner ear possess self-renewal and differentiation capabilities and have been identified as an ideal cellular source for repairing damage to inner ear hair cells (He et al., 2021; Creber et al., 2025). According to the theory of tissue and organ microenvironments, when organ-specific stem cells are transplanted into damaged tissues, they possess the ability to recognize intrinsic signals from the surrounding environment (Jongkamonwiwat et al., 2010; Warnecke et al., 2017; Wobus and Boheler, 2005).

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Subsequently, they may secrete specific cytokines or differentiate into cells with characteristics of the host tissue (Chen et al., 2012; Lopez-Juarez et al., 2019). However, the challenge remains in how to efficiently and safely deliver cells to the inner ear while ensuring their subsequent survival, integration, and functional recovery within the host cochlea (Kros and Steyger, 2019).

This study constructed organoids from in vitro expanded neonatal mouse inner ear progenitor cells and transplanted them into the cochlea of adult mice with kanamycin-induced damage via the posterior semicircular canal route. The aim was to systematically evaluate the survival, engraftment, and restorative effects of transplanted cells on auditory function. We first established a stable combined kanamycin-diuretic ototoxicity model to simulate acute sensorineural hearing loss caused by hair cell loss. Subsequently, combining auditory brainstem response (ABR) testing with immunohistochemical staining, we analyzed the distribution of transplanted cells within the cochlea and the resulting auditory recovery. Results demonstrated that transplanted inner ear progenitor cells effectively survived and extensively colonized the injured cochlea, accompanied by partial restoration of auditory function. These findings provide crucial experimental support for inner ear cell transplantation as a therapeutic strategy for sensorineural hearing loss.

Results

Kanamycin-induced ototoxicity and hair cell loss in adult mice

To establish a stable ototoxic injury model, adult mice received intraperitoneal injections of kanamycin (1 g/kg), followed 30 min later by a diuretic (300 mg/kg, intraperitoneal injection) to synergistically induce ototoxic damage. Auditory threshold changes were assessed via ABR testing prior to drug administration and on day 3 post-modeling. Results showed that mice exhibited a hearing threshold shift of at least 60 dB across all tested frequencies post-injury. At 4 kHz and 32 kHz frequencies, ABR waveforms failed to evoke even with 90 dB stimulation (Fig. 1a). The hearing loss induced by this model remained stable, with no significant recovery observed up to 4 weeks post-injury.

To further validate the injury characteristics morphologically, we performed whole-mount and frozen section immunofluorescence staining on the basilar membranes of the cochlea from normal control and kanamycin-injured mice. Myosin VIIa was used to label surviving hair cells, and DAPI was used to label cell nuclei. Whole-mount staining revealed intact three rows of outer hair cells and one row of inner hair cells in the apical, middle, and basal turns of control cochleae. In contrast, the damaged cochlea exhibited near-complete loss of the three rows of outer hair cells, with only the inner hair cells largely preserved

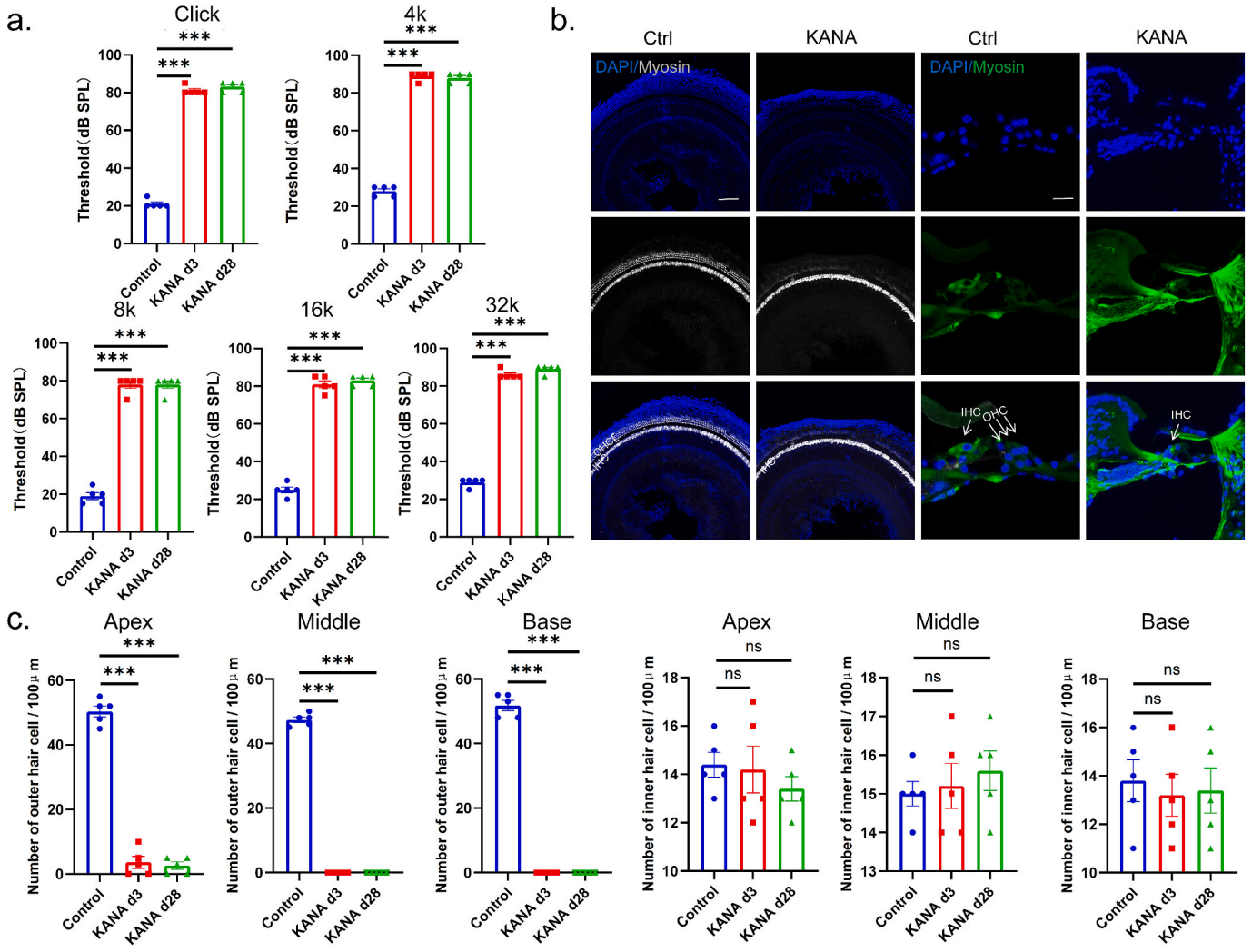


Fig. 1. Kanamycin induces ototoxicity and hair cell loss in adult mice. a. Ototoxic damage induced by kanamycin combined with diuretics leads to elevated hearing thresholds. (n = 5) b. Immunofluorescence images of frozen cochlear sections and whole-mount basilar membrane staining after kanamycin injury reveal extensive loss of outer hair cells in the kanamycin-damaged group compared to the control group. c. Bar charts showing apical, middle, and basal inner hair cells versus outer hair cells in the basilar membrane at days 3 and 28 post-kanamycin model establishment (n = 5). Scale bar = 50 μm.

(Fig. 1b–c). Frozen section staining further confirmed these observations in longitudinal sections: the control group exhibited distinct three columns of outer hair cells and one column of inner hair cells, whereas the damaged group showed extensive loss of outer hair cells with only the inner hair cell column retained.

This study utilizes an aminoglycoside-induced outer hair cell-specific injury model to explore the potential of mouse-derived inner ear stem cell transplantation for structural repair and functional recovery.

Cochlear cell transplantation via the posterior semicircular canal route

The key to successful cell transplantation lies in precisely delivering cells to the inner ear while minimizing surgical trauma and hearing loss. This study employed a posterior semicircular canal injection pathway to transplant cells into the cochlea. As shown in Fig. 2a, the procedure utilized a retroauricular approach: a 1–2 cm incision behind the mouse ear exposed the posterior semicircular canal. A micro-pore was created in the canal wall, through which cells were precisely transplanted into the inner ear. This method induced minimal inflammatory response and did not cause hearing loss in mice. ABR testing was performed preoperatively and at days 7 and 14 postoperatively to assess hearing threshold changes. As shown in Fig. 2b and 2c, no statistically significant differences were observed in hearing threshold measurements between preoperative and postoperative days 7 and 14. This confirms the safety of the surgical approach and indicates no adverse effects on mouse hearing.

Expansion of cochlear progenitor cells in inner ear organoids

The efficacy of cell therapy largely depends on the selection of transplanted cell types and their differentiation status. Previous studies have confirmed the presence of inner ear stem cells with self-renewal and multipotent differentiation potential in both the vestibular system and cochlea (Chen et al., 2026). Under appropriate conditions, these cells can differentiate into hair cell-like structures. For example, certain cells within the adult utricle can autonomously differentiate into hair-cell-like structures even without mitogen stimulation, suggesting their inherent pluripotency. However, the proliferative capacity of endogenous cells within the adult cochlea is limited, and their ability to differentiate into major inner ear cell types is also constrained.

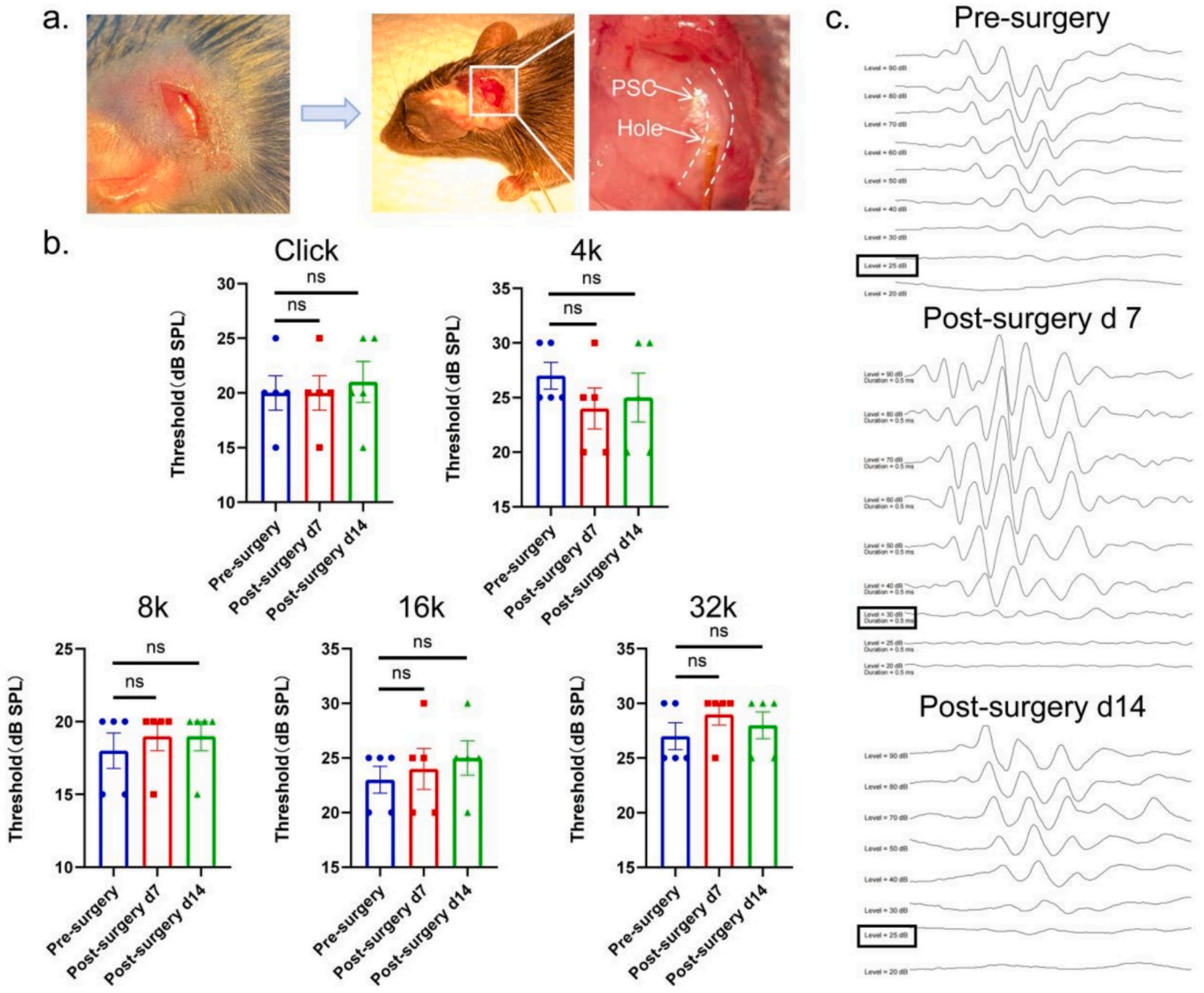


Fig. 2. Cochlear cell transplantation via the posterior semicircular canal route. a. Schematic diagram of the surgical procedure for cell delivery via the posterior semicircular canal route. b. Statistical graphs of auditory brainstem response thresholds at baseline and on postoperative days 7 and 14, showing no statistically significant differences between preoperative and postoperative periods. c. Representative linear graphs of hearing thresholds at baseline and on postoperative days 7 and 14 (n = 5 per group).

Multiple studies have confirmed that Epithelial Cell Adhesion Molecule (EPCAM) positive cell populations isolated from the cochlea of newborn mice exhibit enhanced in vitro expansion capacity, self-renewal potential, and the ability to differentiate into hair cell-like cells and supporting cells (Xu et al., 2024). Therefore, sorting EPCAM-positive cells using flow cytometry represents an effective method for enriching cochlear epithelial progenitor cells. In this study, we employed flow cytometry sorting to isolate EPCAM-positive inner ear progenitor cells from the cochlea of newborn mice (Fig. 3a). These cells were then seeded onto a three-dimensional Matrigel matrix enriched with multiple signaling modulators for in vitro expansion. During culture, the cells gradually aggregated into cell clusters and exhibited a stable expansion trend over time (Fig. 3c). To characterize cells during the expansion phase, we performed immunofluorescence analysis for the expression of antibodies targeting Sox2 (a universal stem cell marker) and Myo6 (an early hair cell marker). Results revealed widespread expression of Sox2 within the organoids, while Myo6 showed only localized weak positivity (Fig. 3b). This indicates that cells in the culture system retain strong stem cell properties while exhibiting early signs of differentiation toward the hair cell lineage.

Dynamic observation revealed that single cells at the initial inoculation stage gradually aggregated into morphologically intact, significantly enlarged three-dimensional cluster structures by days 5, 10, and 14 (Fig. 3c), further confirming that this culture system supports long-term, stable in vitro expansion of inner ear progenitor cells. In summary, the inner ear organoids derived from neonatal mouse cochlear progenitor cells exhibit sustained proliferation and aggregation capacity during the expansion phase while retaining typical stem cell biological characteristics.

Transplantation of cochlear progenitor cells partially restores hearing loss induced by kanamycin

We assessed auditory thresholds in mice using ABR detection. ABR recordings were performed prior to kanamycin-induced damage and 3 days post-damage, with cell transplantation conducted on day 5 post-damage. Additional ABR detections were performed 7 and 14 days after cell transplantation, followed by cochlear extraction for immunofluorescence staining (Fig. 4a). The murine ABR typically elicits five consecutive waves (I-V), each representing neural activity from different stations along the auditory pathway. Specifically, ABR wave I originates from the auditory nerve, reflecting the initial neural response to sound. Waves II-III are generated by the cochlear nucleus and superior olivary complex, indicating neural connections within the central auditory pathway.

Following kanamycin-induced ototoxic injury, auditory function was severely impaired, with hearing thresholds rising from approximately 20 dB SPL to nearly 90 dB SPL. We performed cell transplantation in the left ears of mice with ototoxic damage, with the right ears serving as controls. ABR testing was conducted 14 days after cell transplantation. The cell transplantation group was labeled as transplanted d14, since cell transplantation occurred on day 5 after KANA injury. The control group was labeled as KANA d19. As shown in Fig. 4b, at 14 days post-transplantation, hearing thresholds in the transplanted mice across various frequencies were measured as follows: 38.00 ± 2.00 dB (Click), 34.00 ± 3.67 dB (4 kHz), 36.00 ± 4.00 dB (8 kHz), 46.00 ± 2.92 dB (16 kHz), and 49.00 ± 1.87 dB (32 kHz). In contrast, the thresholds for kanamycin damage in mice on day 19 were: 81.00 ± 1.00 dB (Click), 85.00 ± 1.58 dB (4 kHz), 77.00 ± 2.00 dB (8 kHz), 81.00 ± 1.87 dB (16 kHz), and 86.00 ± 1.87 dB (32 kHz). Compared to the kanamycin-injured group, the cell transplantation group showed significant

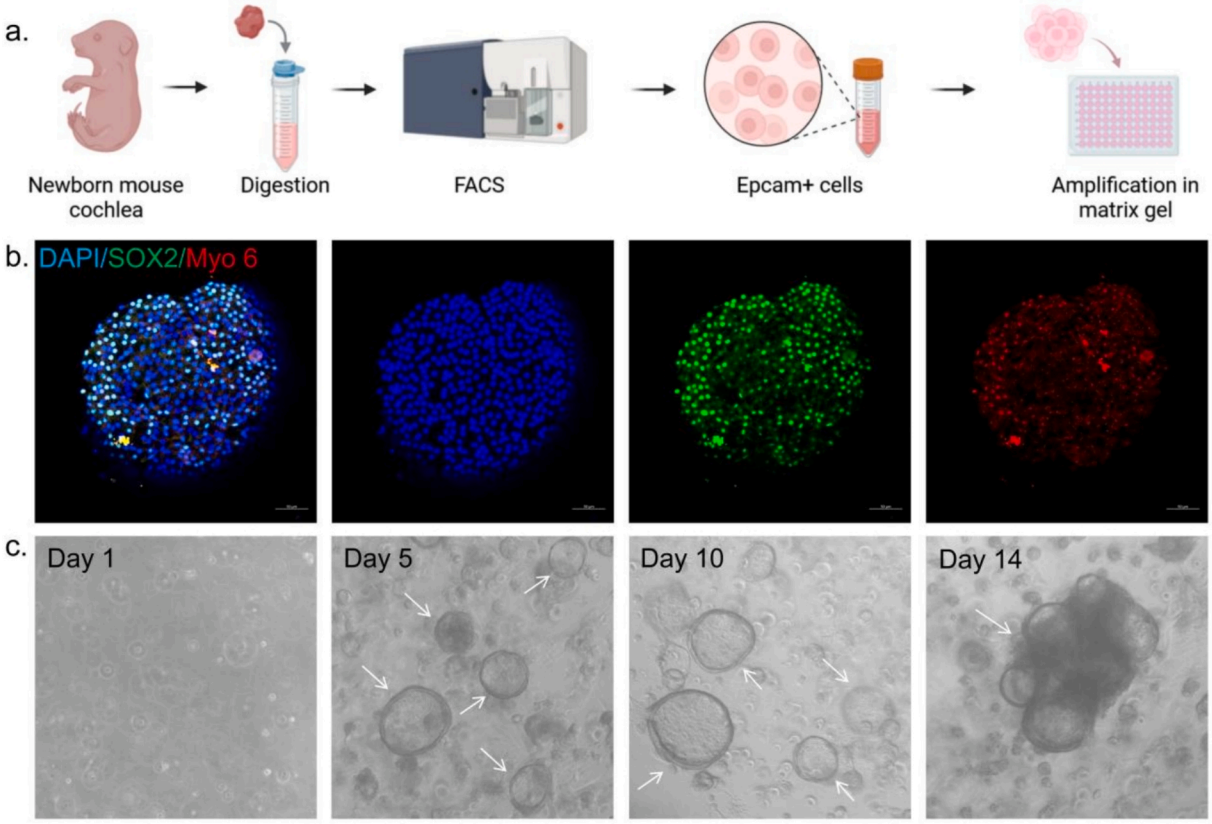


Fig. 3. Expansion of cochlear progenitor cells in inner ear organoids. a. Schematic diagram of the process for constructing inner ear organoids derived from neonatal mice. b. Immunofluorescence staining of inner ear organoids after 14 days of expansion, with Sox2 representing stem cells and Myo 6 representing cochlear hair cells. c. Under light microscopy, inner ear progenitor cells aggregate into cell clusters and stably expand over time. Scale bar = 50 μm.

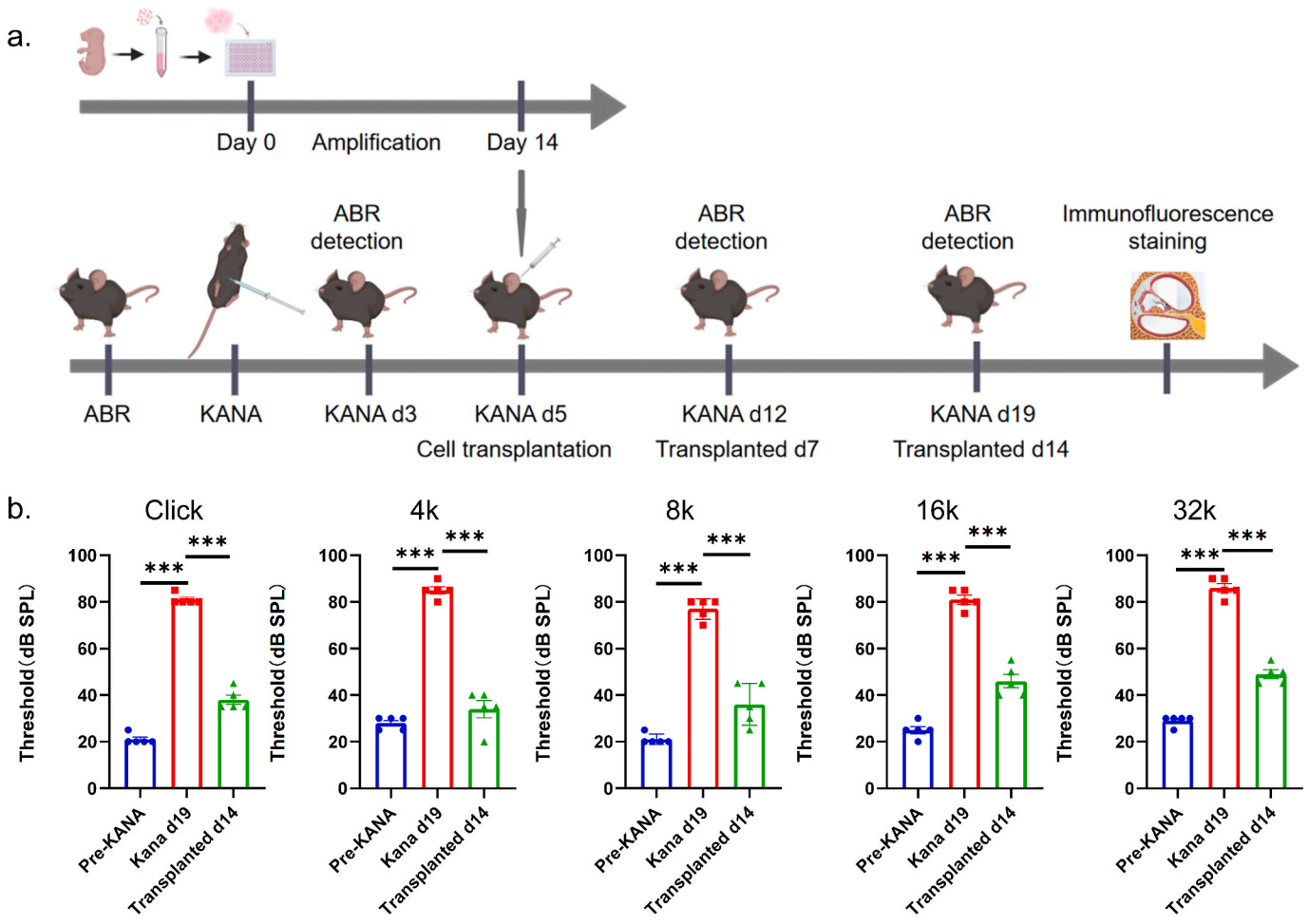


Fig. 4. Cell transplantation partially restores hearing loss induced by kanamycin. a. Schematic diagram of the experimental procedure. b. The bar chart presents statistical analysis results of different frequency hearing thresholds before kanamycin modeling, 19 days after modeling, and 14 days after cell transplantation (n = 5 per group).

recovery in hearing thresholds.

Based on these results, it can be concluded that the in vivo transplantation of cultured cochlear progenitor cells, which possess certain stem-like properties, can partially restore the hearing loss caused by kanamycin-induced ototoxicity.

Distribution of transplanted cells in the ototoxically damaged cochlea

To visualize cell distribution following transplantation into ototoxicity-damaged cochleae, we utilized $SOX2^{CreERT2/+}$; $Rosa26^{tdTomato/+}$ mice with lineage tracing. To achieve SOX2-cell-specific expression of tdTomato in organoid culture and transplantation, we administered tamoxifen to newborn mice to label SOX2-positive cells and their progeny. SOX2 + cells, which possess certain inner ear progenitor cell characteristics (Xiao et al., 2025), were isolated via fluorescence-activated cell sorting (FACS), cultured in vitro for 14 days, and subsequently transplanted in vivo. Fourteen days after transplantation, following ABR assessment, the mouse cochleae were harvested. Immunofluorescence staining was performed on both whole-mount preparations of the cochlear basilar membrane and on cochlear frozen sections.

To accurately trace the localization and distribution of transplanted cells within the inner ear, we labeled cochlear hair cells using Myosin VIIa. Previous studies suggested that the high potassium environment of the endolymph is unfavorable for the survival of transplanted cells within the cochlea. However, our study found that after cells were delivered to the inner ear, a portion of them could migrate toward the scala media. Cochlear frozen section results (Fig. 5a) revealed that tdTomato-labeled transplanted cells primarily localized to the scala tympani, forming sheet-like aggregates beneath the basilar membrane. A small number of cells were also observed distributed in the scala media. Whole-organ cochlear staining (Fig. 5b) further confirmed that transplanted cells could colonize the apical, middle, and basal turns of the cochlea. To more clearly illustrate the final localization of transplanted cells within the inner ear and their relationship with supporting cells and hair cells, we have included a high-magnification immunofluorescence image in Fig. 5c.

In summary, the results obtained from these in vivo cell transplantation experiments indicate that transplanted cochlear progenitor cells can survive and establish themselves within the cochlea.

Discussion

This study first established a kanamycin-induced ototoxic hearing loss model and performed inner ear progenitor cell transplantation via the posterior semicircular canal pathway. Through systematic evaluation of transplanted cell survival, distribution within the damaged cochlea, and their impact on auditory function, we found that transplanted progenitor cells could survive within the host cochlea and primarily colonized the basilar membrane region, with a small number of cells also migrating toward the scala media. More importantly, auditory function was partially restored in the injured mice following transplantation.

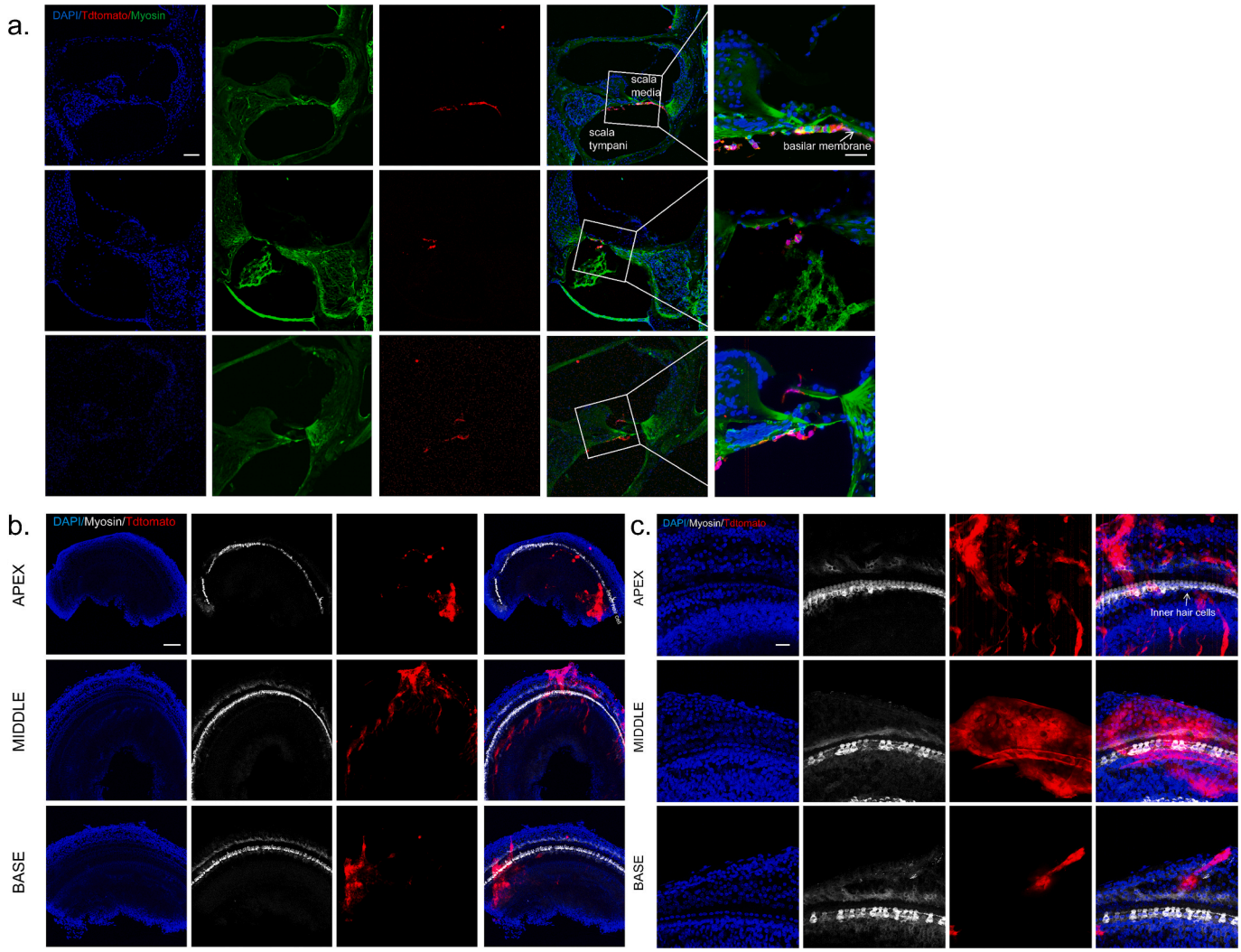


Fig. 5. Distribution of transplanted cells in the ototoxically damaged cochlea. a. Section immunofluorescence staining showing transplanted cells within the mouse cochlea. b. Whole-mount immunofluorescence staining illustrating transplanted cells on the mouse cochlear basilar membrane. c. Enlarged images of cell engraftment on the cochlear basilar membrane under 63 × magnification. Scale bar = 50 μm.

The posterior semicircular canal cochlear implantation procedure has been proven to be a safe and stable surgical approach. This method rarely causes postoperative hearing loss or inflammatory reactions, making it our chosen method for the safe and reliable implementation of inner ear cell transplantation (He et al., 2020). Furthermore, the localization of cells within the scala tympani and their migration toward the scala media indicate that the cochlear microenvironment plays a crucial role in the survival and distribution of transplanted cells. Previous studies have suggested that the hyperkalemic environment of the endolymph is detrimental to cell survival (Kasagi et al., 2013). However, our study found that cells transplanted into the inner ear could colonize in small numbers in the scala media, likely due to acute damage caused by kanamycin altering the inner ear's microenvironment. (Lee et al., 2017; Lang et al., 2008).

Secondly, our investigation unveiled a notable improvement in the auditory function of the mice following transplantation, which is likely due to the influence of the transplanted cells on auditory repair. This influence may be mediated through multiple mechanisms (Peloggia et al., 2021; Chang et al., 2020): 1. Transplanted progenitor cells entering the inner ear may induce the secretion of neurotrophic factors (such as BDNF) as well as inflammation-related cytokines and survival factors to support the function of residual inner hair cells and spiral ganglion neurons. (Kwan et al., 2015; Parker et al., 2007; Zhu et al., 2021; Xu et al., 2024). 2. Transplanted progenitor cells entering the inner ear may differentiate into partially functional hair-like cells under the stimulation of the inner ear microenvironment, or integrate with host supporting cells (Cheng et al., 2025; Li et al.,2025). 3. Transplanting progenitor cells into the inner ear may create a more favorable microenvironment for endogenous repair by modulating the extracellular matrix or enhancing immune cell activity (Chen et al., 2012; Kamiya et al., 2007).

However, this study is not without limitations. First, although hearing thresholds improved post-transplantation, they did not return to normal levels, suggesting room for improvement in cell integration efficiency or functional maturation. Second, as inner ear cell transplantation is an invasive procedure, it cannot be repeated. Consequently, both the number of transplanted cells and their survival duration are constrained. Third, this study primarily focused on establishing the feasibility, safety, and efficacy of achieving functional recovery through cell transplantation. Current data cannot conclusively determine the specific mechanisms underlying these effects, representing a limitation of this study. Future research will employ techniques such as lineage tracing, conditional knockout, and single-cell sequencing to rigorously validate these proposed mechanisms. These shortcomings hinder the clinical application of this therapeutic strategy. Developing delivery vehicles capable of enabling repeated inner ear cell

transplantation will be a prerequisite for large-scale clinical implementation.

In conclusion, this study has provided evidence that transplanting inner ear progenitor cells via the posterior semicircular canal approach can effectively promote the structural and functional recovery of cochleae damaged by kanamycin, offering novel experimental support for cell therapy in sensorineural hearing loss. Future study should focus on optimizing transplantation strategies, enhancing the efficiency of cell-directed differentiation, and exploring the combined use of these cells with neuroprotective agents and biomaterials to accelerate the clinical translation of this therapy.

Experimental methods

Animals and Grouping. Male C57BL/6J mice aged 8 weeks were purchased from Beijing SPF Biotechnology Co., Ltd. (Experimental Animal License No.: SYXK (Jing) 2025–0033. Animal use complied with the Detailed Rules for the Administration of Medical Laboratory Animals and the Capital Medical University Laboratory Animal Management Regulations (AEEI-2024–206). $SOX2^{CreERT2/+}$ mice were obtained from Wan GuoQiang(Model Animal Research Center, MARC), $Rosa26^{tdTomato/+}$ were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. We administered tamoxifen (Sigma, 30 mg/40 g) via intraperitoneal injection for two consecutive days to p0-born $SOX2\text{-}^{CreERT2/+}$; $Rosa26^{tdTomato/+}$ newborn mice, followed by organoid culture at p2. This protocol aims to label cochlear progenitor cells expressing SOX2 at the induction time point and all their progeny. Mice were housed in the Animal Facility of Capital Medical University, with a maximum of 5 mice per cage, ad libitum access to food and water, and a 12 h/12 h light/dark cycle. Animals were randomly assigned to experimental groups using a random number table method.

Kanamycin-induced model establishment. Mice were intraperitoneally injected with kanamycin (Sigma) at a dose of 1 g/kg (10 μl/g of a 100 mg/ml solution). Thirty minutes later, furosemide was administered via intraperitoneal injection at a dose of 300 mg/kg (30 μl/g of a 10 mg/ml solution). Three days after administration, changes in the auditory thresholds and hair cells of the mice were examined.

Auditory Brainstem Response (ABR) Measurement. Animals were weighed prior to anesthesia. Anesthesia was induced via intraperitoneal injection of ketamine (100 mg/kg, Gutian Pharmaceutical Co., Ltd., Fujian, China) and xylazine (10 mg/kg, Sigma-Aldrich Co., Llc., United States). Electrode tips were cleaned with alcohol before placement. The recording electrode was positioned subcutaneously at the midline of the forehead, the reference electrode subcutaneously behind the test ear, and the ground electrode subcutaneously behind the non-test ear. The external speaker was placed approximately 1–2 cm from the test ear canal. Testing was performed using a TDT RZ6 system and BioSigRZ (Tucker Davis Technologies) software. Stimuli included clicks and tone bursts at frequencies of 4 kHz, 8 kHz, 16 kHz, and 32 kHz. Recording commenced at 90 dB SPL, decreasing in 10 dB steps, and switching to 5 dB steps near threshold. The threshold was defined as the lowest intensity producing a repeatable ABR waveform (confirmed twice). ABR thresholds at Click, 4 kHz, 8 kHz, 16 kHz, and 32 kHz were recorded.

Semicircular Canal Surgery Five days post-modeling in Kanamycin-treated mice, weigh the animals and anesthetize them via intraperitoneal injection of ketamine (100 mg/kg) and thiopental (10 mg/kg). The posterior semicircular canal is exposed through a post-auricular approach. A small fenestration is created on the posterior semicircular canal using a 26-gauge needle, which is then gently rotated to achieve an appropriate aperture size. Cells were re-suspended in DMEM at the density of 1×10^4 cells per mL. We injected approximately 2 μl of the cell suspension into the inner ear.The cell suspension was inoculated into the posterior semicircular canal at a rate of 0.5 μl/min using a micro injection pump. After injection, the fenestration is sealed with muscle tissue, and the surgical wound is closed using No. 4 sutures.

In vitro organoid culture. The cochlear basilar membranes were dissected from neonatal mice and digested with 0.125% trypsin (Thermo Fisher Scientific) in a water bath at 37 °C for 30 min. Digestion was terminated by adding 3 mL of 10% FBS. The cell suspension was filtered through a 40 μm cell strainer, centrifuged, and the supernatant was discarded. Cells were resuspended in a solution containing EPCAM antibody and incubated for 30 min. After washing away unbound antibodies 2–3 times, cells were sorted using fluorescence-activated cell sorting (FACS).

Primary cochlear progenitor cell culture and expansion. Sorted cells were resuspended in Matrigel and the mixture was plated onto culture plates. Expansion medium was added to support proliferation. The expansion medium consisted of DMEM/F-12 supplemented with penicillin/streptomycin, N2, and B27 (all from Thermo Fisher Scientific), 50 ng/mL EGF (Peprotech), 50 ng/mL bFGF (Peprotech), 50 ng/mL IGF (Peprotech), and 3 μM CHIR99021 (Sigma). Organoids were passaged every 8–10 days by mechanically pipetting them into small fragments at a split ratio of 1:6 to 1:8.

Cryosectioning. The mouse cochleae were dissected and fixed in 4% PFA. Subsequently, they were decalcified in EDTA solution at room temperature for 2–3 days, followed by overnight immersion in 30% sucrose solution. Afterward, the samples were embedded in OCT compound, sectioned, mounted onto adhesive slides, and dried in an incubator for subsequent immunofluorescence staining and confocal microscopy observation.

Immunofluorescence Staining. Under deep anesthesia (ketamine 100 mg/kg, xylazine 10 mg/kg), animals were euthanized by cervical dislocation. Heads were removed, cochleae carefully dissected under a stereomicroscope, and fixed overnight at 4 °C in phosphate-buffered saline (PBS) containing 4% paraformaldehyde. Fixed cochleae were decalcified in 10% EDTA for 1–2 h. The basilar membrane segments (apical, middle, basal turns) were dissected in PBS, removing the tectorial membrane and Reissner's membrane. Samples were permeabilized with 0.3% Triton X-100 (Sigma-Aldrich) in PBS for 30 min at room temperature (RT), then blocked with 5% goat serum (ZSGB-BIO, Beijing, China) for 1 h at RT. Primary antibodies were applied and samples incubated overnight at 4 °C: rabbit antimyosin-VIIa antibody (1:300, Proteus BioSciences Inc., 256790). The dilution buffer for the primary antibody used in immunofluorescence staining is 0.3% Triton X-100 and 5% goat serum. The following day, samples were washed three times in PBS (10 min each). Secondary antibodies were applied and incubated for 2 h at RT: goat antirabbit IgG (H + L) Alexa Fluor 647/488, goat anti-mouse IgG1 Alexa Fluor 568 (1:300, Invitrogen, A-21245, A-21124, respectively). After three additional PBS washes (10 min each), samples were mounted on slides using mounting medium containing DAPI (ZSGB-BIO).

Confocal Microscopy Imaging. Images were acquired using a 63x oil immersion high-resolution confocal microscope (TCS SP8 II; Leica Microsystems). Z-stack images were obtained by scanning from top to bottom with a 0.5 μm step size. Samples were observed at optimal excitation wavelengths: 488 nm (green), 568 nm (red), 647 nm (silver), and 358 nm (blue) for DAPI visualization.

Statistical Analysis All data are presented as mean ± standard error of the mean (SEM), with experiments repeated at least three times. Statistical analysis was performed using GraphPad Prism 8.0.2 (GraphPad Software Inc., La Jolla, CA, United States). Intergroup statistical differences were assessed using two-way analysis of variance (ANOVA) followed by multiple comparisons tests. In all analyses, $p < 0.05$ was considered statistically significant. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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CRediT authorship contribution statement

Shan Liu: Writing – original draft, Validation, Investigation, Data curation. **Rui Li:** Methodology, Investigation. **Fei Wang:** Writing – review & editing. **Yan Wang:** Visualization, Data curation. **Ke Liu:** Supervision, Project administration, Funding acquisition, Conceptualization.

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